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DECLARATION OF AUTHENTICITY

Assignment:	KM02 - Logical Thinking and Basic Calculations. Learner Workbook
Date Submitted:	26 October 2025
Qualification / Skills Programme:	Occupational Certificate: Data Science Practitioner | SAQA ID: 118708
Assessment Site / Provider:	Praesignis PTY LTD | MICT SETA

Full Name:	Khensani Hlongwane
ID Number:	0102110783085
Student Number:	hlongwanekhensani678@gmail.com

I, the above-mentioned learner, hereby declare that:

- The attached submission is my own original work.
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Learner Signature: **Date:** 26 October 2025

Assignment Portfolio

Student Information

Name: Khensani Hlongwane

Student ID: hlongwanekhensani678@gmail.com

Email: hlongwanekhensani678@gmail.com

Assignment Details

Assignment: KM02 - Logical Thinking and Basic Calculations. Learner Workbook

Submission Date: Sunday, 26 October 2025, 10:36 AM

Grade: 97.20 / 100.00 (97.2%)

Graded: Friday, 6 March 2026, 2:47 PM

Submitted Files

- KM-02 Learner workbook.pdf

Marking Guide Assessment

Final score: 139.00 / 143.00 (97.20%)

KM-02-KT01: Mathematical Thinking Skills

Description: Explain how mathematical thinking improves reasoning and decision-making in data science.

Markers: Answer Guide & Mark Allocation:
(2 marks) For explaining that it provides a structured, logical framework for breaking down complex problems into smaller, manageable steps.
(2 marks) For stating that it emphasizes precision, evidence, and quantifiable metrics, moving decisions from "gut feeling" to data-driven conclusions.
(1 mark) For mentioning that it helps in identifying patterns, relationships, and anomalies within data.
Facilitator Notes: Look for answers that connect abstract mathematical logic to the practical process of data analysis.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT02: Decimal and Binary Systems

Description: Explain how the binary system differs from decimal. Use 10011_2

Markers: (2 marks) Difference: Decimal is base-10 (digits 0-9), Binary is base-2 (digits 0-1).

(3 marks) Explanation of 10011_2:

$1 * 2^4 = 16$
$0 * 2^3 = 0$
$0 * 2^2 = 0$
$1 * 2^1 = 2$
$1 * 2^0 = 1$
Total: 19_{10}

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT02: Decimal and Binary Systems

Description: Convert decimal 25 to binary

Markers: Answer Guide (Repeated Division):

25 / 2 = 12 R 1 (LSB - Least Significant Bit)
12 / 2 = 6 R 0
6 / 2 = 3 R 0
3 / 2 = 1 R 1
1 / 2 = 0 R 1 (MSB - Most Significant Bit)
(5 marks) for correct answer: 11001_2. Award partial marks for correct method but wrong final answer.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT02: Decimal and Binary Systems

Description: Add 1011_2 and 1101_2.

Markers: Answer Guide:

```
text
 1011
+ 1101
------
 11000
```

(5 marks) for correct answer: 11000_2. Award partial marks for correct addition with a minor carry-over error.

Score: **4.00 / 5.00 (80.00%)**

KM-02-KT03: Size and Magnitude

Description: List five prefixes from Giga to Pico.

Markers: (5 marks - 1 mark per correct prefix/symbol/power)

Answer Guide:
Prefix
Symbol
Power of Ten
Giga
G
10^9
Mega
M
10^6
Kilo
k
10^3
Milli
m
10^{-3}
Micro
µ
10^{-6}
Nano
n
10^{-9}
Pico
p
10^{-12}

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT03: Size and Magnitude

Description: Convert the values

Markers: (6 marks - 1 mark each)
Answer Guide:
a. 3,200,000 bps = 3.2 Mbps (÷ 1,000,000)
b. 0.000000000045 s = 45 ps (× 10^{12})
c. 0.00089 W = 890 µW (× 1,000,000)
d. 5,400,000 bytes = 5.4 MB (÷ 1,000,000)
e. 0.000003 A = 3 µA (× 1,000,000)
f. 1,500,000,000 bytes = 1.5 GB (÷ 1,000,000,000)

Score: **5.00 / 6.00 (83.33%)**

KM-02-KT04: Error in Calculations

Description: What identifies a number as rational or irrational?

Markers: Answer Guide:
(3 marks) A rational number can be expressed as a fraction (ratio) of two integers (e.g., 1/2, 0.75, 5).
(2 marks) An irrational number cannot be expressed as a fraction of two integers. Its decimal representation is non-terminating and non-repeating (e.g., π, √2).

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT04: Error in calculations

Description: How is the value of an expression determined?

Markers: The value of an expression is determined by evaluating its components according to a fixed set of rules, primarily:

Operator Precedence: The order in which operations are performed (e.g., multiplication before addition, as defined by PEDMAS/BODMAS rules).

Operator Associativity: The direction (left-to-right or right-to-left) in which operations of the same precedence are evaluated.

Data Types: The types of values involved (e.g., integers, floats) can influence the result's type and value, especially when mixing types (e.g., integer vs. float division).

Parentheses: Expressions within parentheses ( ) are always evaluated first, overriding the standard precedence

rules.

This rule-based system ensures the expression is evaluated consistently, unambiguously, and accurately every time, both in mathematics and in programming.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT04: Error in Calculations

Description: Self-Check Quiz
1. What is the correct order of operations used to evaluate a mathematical expression?
2. Evaluate the expression: 8 + 4 * 2
3. Evaluate the expression: (8 + 4) * 2
4. Why is it important to follow the correct order of operations when simplifying an expression?
5. What is the value of this expression? 10 - 4 + 2

Markers: Self-Check Quiz Answers:(5 marks = 1 mark each)
1. C
2. B (8 + 8 = 16)
3. A ((12) * 2 = 24)
4. C
5. B (10 - 4 = 6, then 6 + 2 = 8)

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT05: Cartesian Coordinate System

Description: Identify and explain the components.

Markers: Answer Guide:
• (2 marks) Axes: The two perpendicular lines: the x-axis (horizontal) and y-axis (vertical).
• (2 marks) Origin: The point (0, 0) where the x-axis and y-axis intersect.
• (1 mark) Coordinates: A pair (x, y) used to locate any point on the plane.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT05: Cartesian coordinate system

Description: Define each of the four quadrants in a Cartesian plane. What are the sign conventions for x and y in each?

Markers: The Four Quadrants of the Cartesian Plane
The Cartesian plane is divided into four sections called quadrants, which are numbered counterclockwise starting from the top-right.
Quadrant I (Top-Right)
x-coordinate: Positive (+)
y-coordinate: Positive (+)
Sign Convention: (+, +)

Quadrant II (Top-Left)
x-coordinate: Negative (-)
y-coordinate: Positive (+)
Sign Convention: (-, +)

Quadrant III (Bottom-Left)
x-coordinate: Negative (-)
y-coordinate: Negative (-)
Sign Convention: (-, -)

Quadrant IV (Bottom-Right)
x-coordinate: Positive (+)
y-coordinate: Negative (-)
Sign Convention: (+, -)

Score: **4.00 / 5.00 (80.00%)**

KM-02-KT05: Cartesian Coordinate System

Description: Self-Check Quiz: The Cartesian Coordinate System
1. What are the two axes in a Cartesian coordinate system?
2. Where is the origin located in the Cartesian coordinate system?
3. Which of the following correctly describes Quadrant II in a Cartesian plane?
4. In which quadrant are both x and y values negative?

5. What is the correct order of quadrants in a standard Cartesian plane (moving counterclockwise from the top right)?
Markers: Self-Check Quiz Answers:(5 marks = 1 mark each)
1. B
2. B
3. B
4. C
5. A

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT06: Pythagorean Theorem

Description: : Explain the concept of a Cartesian grid plan and its purpose.

Markers: Answer Guide:
• (3 marks) It is a 2D plane formed by two perpendicular number lines (axes), creating a grid of unique coordinate points.
• (2 marks) Its purpose is to visually represent and analyze relationships between two variables or to locate points in a 2D space.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT06: Pythagorean Theorem

Description: Self-Check Quiz: Pythagorean Theorem & Distance on the Cartesian Grid
1. What does the Pythagorean Theorem calculate in a right-angled triangle?
2. Which of the following is the correct distance formula between two points on a Cartesian plane?
3. What are the “legs” of the triangle when applying the Pythagorean Theorem to two points on a grid?
4. Given two points A(2, 3) and B(6, 7), what is the distance between them?
5. Why is the Pythagorean Theorem useful in data science?
Markers: Self-Check Quiz Answers: (5 marks = 1 mark each)
1. B
2. C
3. C
4. C ($\sqrt{(6-2)^2 + (7-3)^2} = \sqrt{16 + 16} = \sqrt{32}$)
5. C

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT07: Operator Precedence

Description: What is operator precedence? Provide an example.

Markers: Answer Guide:
• (2 marks) Definition: The set of rules that determines the order in which operations are evaluated in an expression.
• (3 marks) Example: In 3 + 4 * 5, multiplication has higher precedence than addition, so it's 3 + (4*5) = 23, not (3+4)*5 = 35.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT07: Operator Precedence

Description: Why is it important?

Markers: Answer Guide:
• To ensure that expressions are evaluated consistently and accurately by everyone (and by computers), eliminating ambiguity.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT07: Operator Precedence

Description: Self-Check Quiz: Operator Precedence
1. Which term best describes "operator precedence"?
2. Why is operator precedence important in both math and programming?
3. Which of the following expressions evaluates to 14?
4. In the expression: 8 + 4 * 2 - 3
What is the correct order of operations?
5. According to PEDMAS, which operation should be performed first?
6. What is the result of this expression in most programming languages?

5 + 6 * 2
7. Which of the following best describes how PEDMAS is applied in programming?

Markers: Self-Check Quiz Answers:(7 marks = 1 mark each)
1. B
2. C
3. B (A=14, B=20, C=14, D=10)
4. B
5. C
6. B (5 + 12 = 17)
7. C

Score: **6.00 / 7.00 (85.71%)**

KM-02-KT08: Integer Division

Description: What role does division play in data analysis?

Markers: Answer Guide: It is used for normalization, calculating averages, rates, proportions, and percentages, which are fundamental to understanding and comparing data.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT08: Integer Division

Description: Divide 10 by 4 and explain integer division.

Markers: • (2 marks) 10 / 4 = 2.5
• (3 marks) Integer division (e.g., 10 // 4 in Python) discards the fractional part (the remainder) and returns only the whole number quotient, which is 2.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT08: Integer Division

Description: What is the rule for dividing integers regarding signs?

Markers: Answer Guide: The sign of the result follows standard multiplication rules:
• (+)/(+) = +
• (-)/(-) = +
• (+)/(-) = -
• (-)/(+) = -

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT08: Integer Division

Description: Self-Check Quiz: Dividing Integers
1. What is the result of this division?
2. Which of the following is the correct result of dividing a negative number by a positive number? -15 ÷ 3
3. What is the sign of the result when you divide two integers with the same sign?
4. What is the result of this operation? -24 ÷ -6
5. Which of the following statements is TRUE?
Markers: Self-Check Quiz Answers: (5 marks = 1 mark each)
1. B
2. A
3. B
4. B
5. B

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT09: Modulus

Description: Explain how the modulus rule is applied in 17 % 5.

Markers: Answer Guide:
• (2 marks) The modulus operator (%) returns the remainder after integer division.
• (3 marks) 17 ÷ 5 = 3 with a remainder of 2. Therefore, 17 % 5 = 2.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT09: Modulus

Description: Self-Check Quiz: Application of the Modulus Operator (%)
1. What is the result of the following operation?
10 % 3
2. What does the expression 15 % 5 evaluate to?
3. Which use case is appropriate for applying the modulus operator?
4. What is the result of -13 % 5 in most programming languages?
5. In data science, how might the modulus operator be practically applied?
Markers: Self-Check Quiz (Modulus) Answers:
1. A
2. A
3. B
4. A (Explanation: It depends on the language, but -13 % 5 commonly returns -3 as it preserves the sign of the dividend. However, some languages return 2. "A" is the most common answer.)
5. C

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT09: Modulus

Description: Self-Check Quiz: Application of the Equality Operator (==)
1. What does the operator == do in programming?
2. What is the result of this expression?
7 == 5 + 2
3. In Python, what will be the output of this code?
a = 10
b = 20
a == b
4. Which of the following comparisons will return True?
5. How is the == operator commonly used in data science programming?
Markers: Self-Check Quiz (Equality) Answers: (5 marks = 1 mark each)
1. B
2. A
3. D
4. B (In Python, 3 and 3.0 are considered equal by ==)
5. B

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT10: Increments

Description: Explain increments with an example.

Markers: Answer Guide:
• (2 marks) Definition: Increasing a variable's value by a fixed amount, usually 1.
• (3 marks) Example: x = 5; x = x + 1; Now x is 6. The shorthand is x++ or x += 1.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT10: Increments

Description: What are compound assignment operators? With an example.

Markers: Answer Guide:
• (2 marks) Definition: Operators that perform an operation and an assignment in one step.
• (3 marks) Example: +=, -=, *=, /=. E.g., count += 5 is shorthand for count = count + 5.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT10: Increments

Description: Use y *= 3 to explain the concept.

Markers: Answer Guide:
• This means y = y * 3.
• It takes the current value of y, multiplies it by 3, and then assigns the result back to the variable y.

Score: **5.00 / 5.00 (100.00%)**

KM-02-KT11: Mixing Types

Description: Self-Check Quiz: Mixing Types and Expression Evaluation

1. What is the result and type of the following expression?
5 + 3.0 * 2
2. Which expression gives a different result than the others?
3. What will be the type of the result in this expression?
result = 7 / 2
4. Which of the following best explains what the computer does in the expression x += 2.5, where x is initially an integer?
5. What will the final value of x be in the following?
x = 5
x *= 2.0
Markers: Self-Check Quiz Answers: (5 marks = 1 mark each)
1. B (3.0 * 2 = 6.0, then 5 + 6.0 = 11.0. Mixing int and float results in a float.)
2. B (A & C = 12.0, B = 7.0)
3. B (The / operator in Python always returns a float, even if the result is a whole number.)
4. C (The variable will be promoted to a float to accommodate the fractional part.)
5. B (x becomes 5 * 2.0 = 10.0, a float.)

Score: **5.00 / 5.00 (100.00%)**

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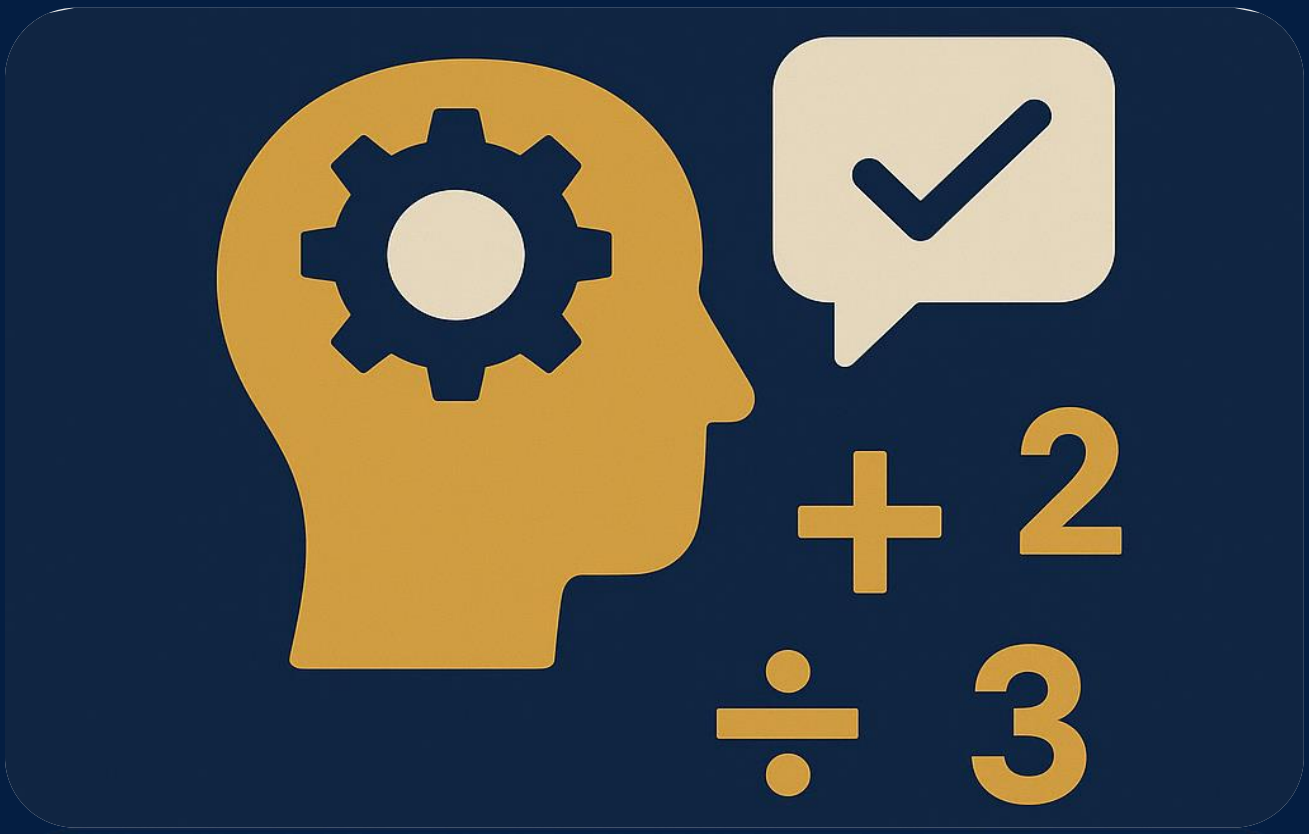
KNOWLEDGE MODULE-02: LOGICAL THINKING AND BASIC CALCULATIONS - REFRESHER



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MATHEMATICAL THINKING SKILLS FOR PROBLEM SOLVING

KM-02-KT01: Mathematical thinking skills for problem solving

1. **Explain how mathematical thinking improves reasoning and decision-making in data science.**

Mathematical thinking provides a structured, logical framework for problem-solving. It improves reasoning by encouraging clarity, precision, and the breakdown of complex problems into smaller, manageable steps. This leads to better decision-making in data science by allowing for the formulation of hypotheses, the creation of quantitative models, and the rigorous validation of results, thereby reducing bias and ensuring conclusions are based on evidence and sound logic rather than intuition alone.

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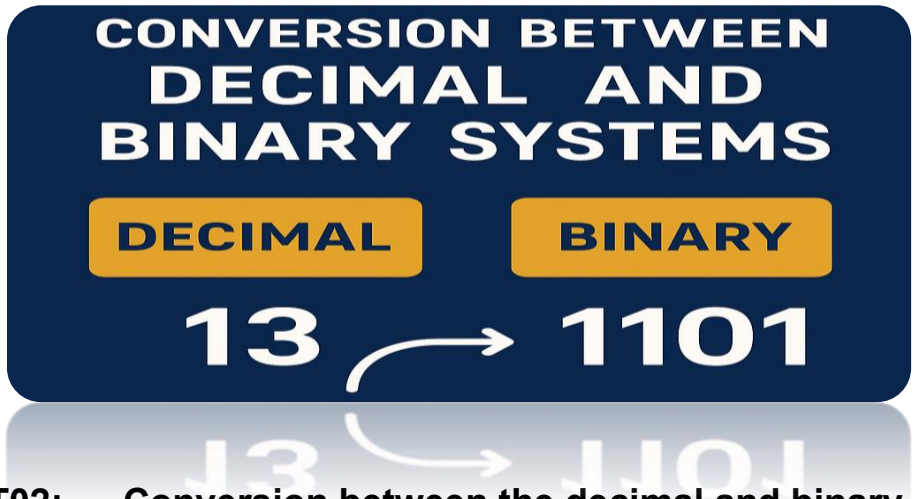

KM-02-KT02: Conversion between the decimal and binary systems

1. **Explain how the binary number system differs from the decimal system. Use the number 10011_2 to explain the value each digit contributes to the overall number.**

1. The decimal system is a base-10 system, using digits 0-9, where each place value is a power of 10. The binary system is a base-2 system, using only digits 0 and 1, where each place value is a power of 2.
   For 10011_2:
2. The rightmost digit (1) is the 2^0 place, contributing 1. ✓

   The next digit (1) is the 2^1 place, contributing 2. ✓

   The next digit (0) is the 2^2 place, contributing 0.

   The next digit (0) is the 2^3 place, contributing 0. ✓

   The leftmost digit (1) is the 2^4 place, contributing 16. ✓
   So, $10011_2 = 16 + 0 + 0 + 2 + 1 = 19_{10}$. ✓

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2.Convert the decimal number 25 to binary. Show all your working steps.

Convert the decimal number 25 to binary. Show all your working steps.

25 ÷ 2 = 12 remainder **1** (LSB - Least Significant Bit)
12 ÷ 2 = 6 remainder **0**
6 ÷ 2 = 3 remainder **0**
3 ÷ 2 = 1 remainder **1**
1 ÷ 2 = 0 remainder **1** (MSB - Most Significant Bit)
= $\mathbf{11001_2}$.

3. Add the binary numbers 1011 and 1101.

1011
+ 1101

11000

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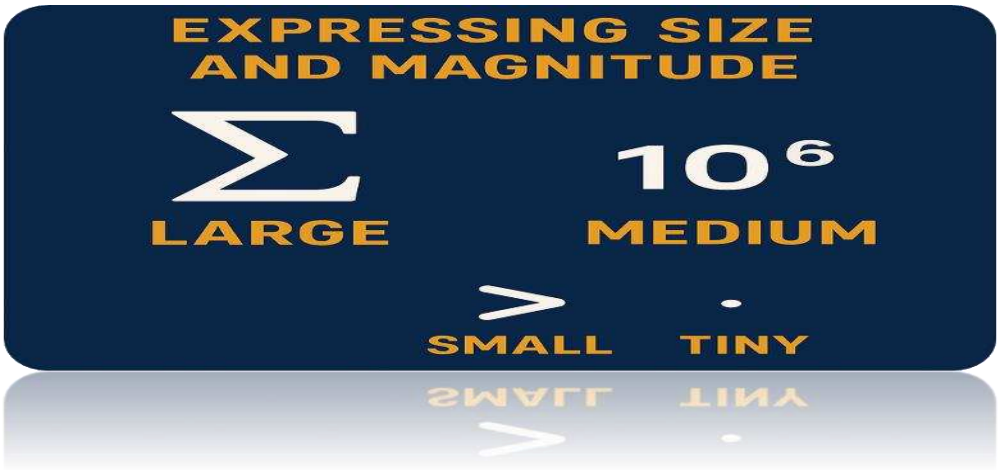

KM-02-KT03: Expressing size and magnitude

1. **List five prefixes from Giga to Pico, their symbols, and powers of ten.**

- Giga (G), 10^9 ✓
- Mega (M), 10^6 ✓
- Kilo (k), 10^3 ✓
- Milli (m), 10^{-3} ✓
- Micro (μ), 10^{-6} ✓
- Pico (p), 10^{-12}

2. **Convert the following values to the unit specified in brackets.**
   a. 3,200,000 bits per second (to Mbps)
   b. 0.000000000045 seconds (to ps)
   c. 0.00089 watts (to μW)
   d. 5,400,000 bytes (to MB)

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e. 0.000003 amperes (to μA)

f. 1,500,000,000 bytes (to GB)

a. 3,200,000 bits per second = **3.2 Mbps** (÷ 1,000,000) ✓
b. 0.000000000045 seconds = **0.045 ps** (× 1,000,000,000,000) ✓
c. 0.00089 watts = **890 μW** (× 1,000,000) ✓
d. 5,400,000 bytes = **5.4 MB** (÷ 1,000,000) ✓
e. 0.000003 amperes = **3 μA** (× 1,000,000) ✓
f. 1,500,000,000 bytes = **1.5 GB** (÷ 1,000,000,000) ✓

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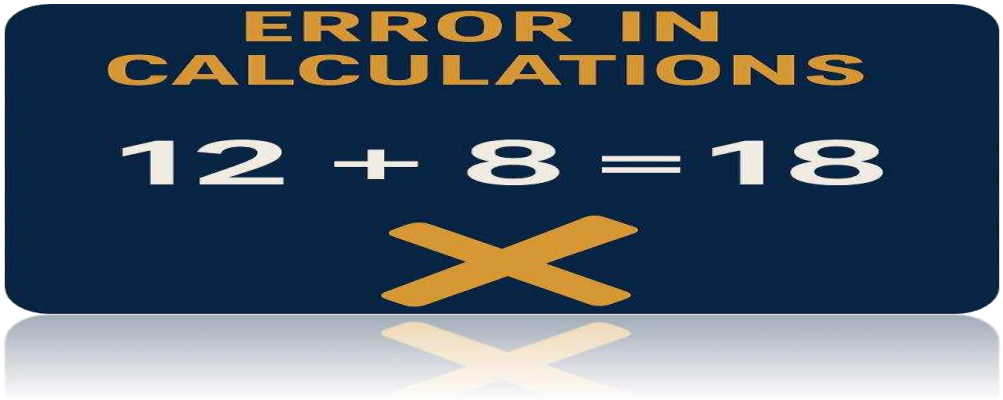

KM-02-KT04: Error in calculations

1. **What properties identify a number as rational or irrational?**

A **rational number** can be expressed as a fraction p/q where p and q are integers and q ≠ 0. They have decimal expansions that either terminate or repeat.
An **irrational number** cannot be expressed as such a fraction. Its decimal expansion is non-terminating and non-repeating.

2. **How is the value of an expression determined?**

The value of an expression is determined by evaluating it according to a fixed set of rules, primarily the **order of operations** (PEMDAS/BODMAS), which ensures consistency and accuracy.

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Self-check quiz

Instructions: Highlight the most appropriate answer for each question.

1. **What is the correct order of operations used to evaluate a mathematical expression?**

   A. Addition, Subtraction, Multiplication, Division

   B. Multiplication, Division, Addition, Subtraction

   C. Parentheses, Exponents, Multiplication & Division (left to right), Addition & Subtraction (left to right) ✓

   D. Left to right, regardless of operator

2. **Evaluate the expression:**

   8 + 4 * 2

   A. 24

   B. 16 ✓

   C. 20

   D. 8

3. **Evaluate the expression:**

   (8 + 4) * 2

   A. 24 ✓

   B. 16

   C. 20

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D. 12

4. **Why is it important to follow the correct order of operations when simplifying an expression?**

A. To reduce the number of steps

B. To match calculator syntax

C. To ensure the expression is evaluated accurately and consistently

D. It's only important in programming

5. **What is the value of this expression?**

10 - 4 + 2

A. 4

B. 8

C. 6

D. 2

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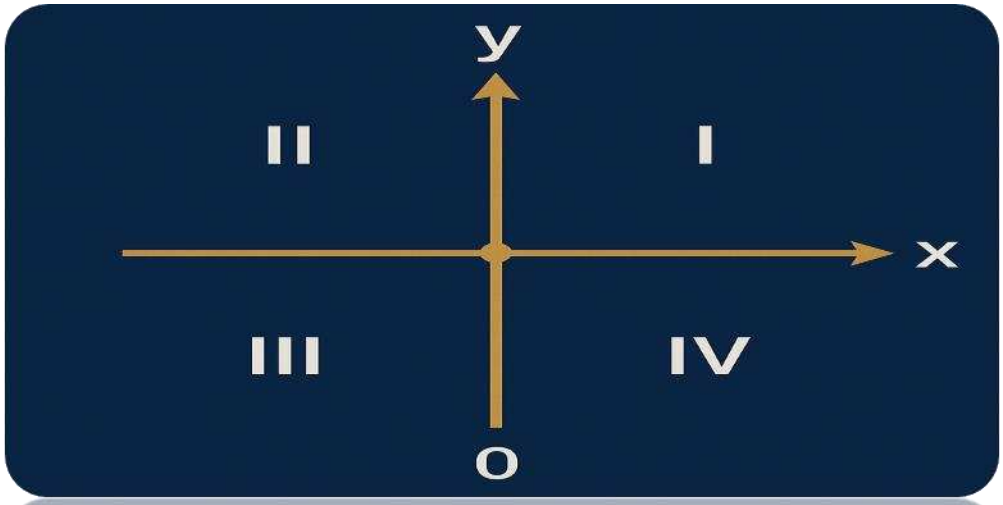

KM-02-KT05: Cartesian coordinate system

1. **Identify and explain the components of the Cartesian coordinate system, including axes and the origin.**

The Cartesian coordinate system is defined by two perpendicular number lines: the **x-axis** (horizontal) and the **y-axis** (vertical). The point where these two axes intersect is called the **origin**, which has coordinates (0, 0). This system is used to uniquely locate any point in a 2D plane using an ordered pair (x, y).

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2. **Define each of the four quadrants in a Cartesian plane. What are the sign conventions for x and y in each?**

- **Quadrant I**: Top-right. (x > 0, y > 0) ✓
- **Quadrant II**: Top-left. (x < 0, y > 0) ✓
- **Quadrant III**: Bottom-left. (x < 0, y < 0) ✓
- **Quadrant IV**: Bottom-right. (x > 0, y < 0) ✓

Self-Check Quiz: The Cartesian Coordinate System

Instructions: Highlight the best answer for each question. This quiz helps reinforce your understanding of the structure and components of the Cartesian coordinate system.

1. **What are the two axes in a Cartesian coordinate system?**

   A. Horizontal and vertical lines with no labels

   B. x-axis (horizontal) and y-axis (vertical) ✓

   C. Axis A and Axis B

   D. Positive and negative lines

2. **Where is the origin located in the Cartesian coordinate system?**

   A. At the top-right of the grid

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B. At coordinates (0, 0), where the x-axis and y-axis intersect ✓

C. At coordinates (1, 1)

D. At the center of Quadrant I

3. **Which of the following correctly describes Quadrant II in a Cartesian plane?**

A. $x > 0, y > 0$

B. $x < 0, y > 0$ ✓

C. $x < 0, y < 0$

D. $x > 0, y < 0$

4. **In which quadrant are both x and y values negative?**

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV ✗

5. **What is the correct order of quadrants in a standard Cartesian plane (moving counterclockwise from the top right)?**

A. I, II, III, IV ✓

B. I, IV, III, II

C. II, I, IV, III

D. IV, I, II, III

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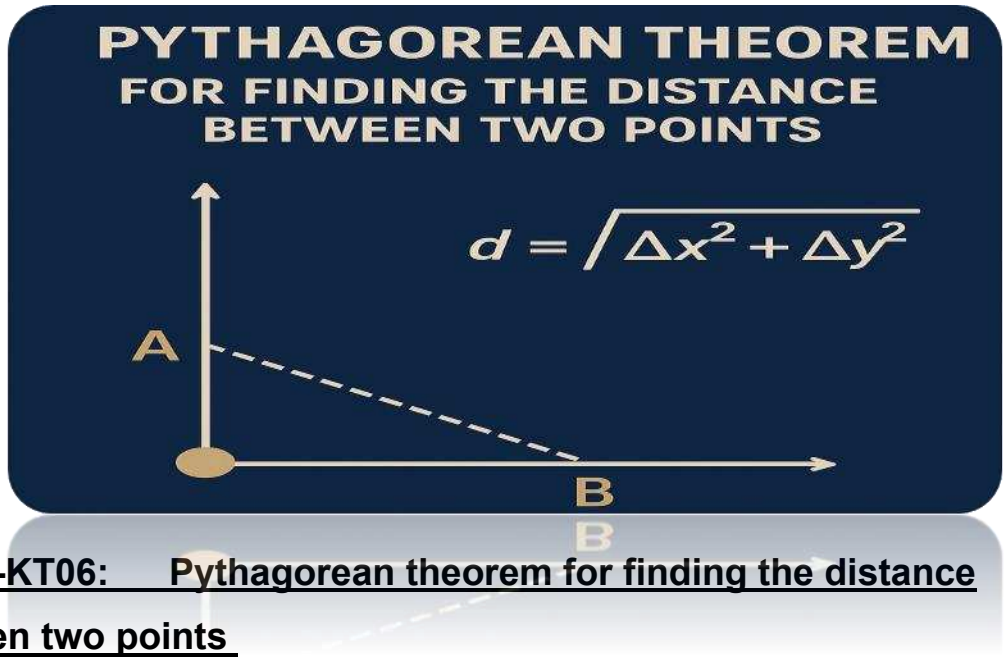

KM-02-KT06: Pythagorean theorem for finding the distance between two points

1. **Explain the concept of a Cartesian grid plan and its purpose**

A Cartesian grid plan is a network of horizontal and vertical lines that form squares, using the Cartesian coordinate system. Its purpose is to provide a precise and systematic way to locate points, represent geometric shapes, and visualize relationships between numerical values in a two-dimensional space, which is fundamental in fields like data visualization, mapping, and graphics.

Self-Check Quiz: Pythagorean Theorem & Distance on the Cartesian Grid

Instructions: Highlight the most appropriate answer for each question. Show working where applicable.

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1. **What does the Pythagorean Theorem calculate in a right-angled triangle?**

   A. The angle between two sides

   B. The length of the hypotenuse ✓

   C. The slope of the triangle

   D. The perimeter of the triangle

2. **Which of the following is the correct distance formula between two points on a Cartesian plane?**

   A. (x1−x2)+(y1−y2)$(x_1 - x_2) + (y_1 - y_2)$

   B. (x1+x2)2+(y1+y2)2\sqrt{$(x_1 + x_2)$^2 + $(y_1 + y_2)$^2}

   C. (x2−x1)2+(y2−y1)2\sqrt{$(x_2 - x_1)$^2 + $(y_2 - y_1)$^2} ✓

   D. |x2−x1|+|y2−y1|$|x_2 - x_1| + |y_2 - y_1|$

3. **What are the "legs" of the triangle when applying the Pythagorean Theorem to two points on a grid?**

   A. The actual points themselves

   B. The x and y axes

   C. The horizontal and vertical distances between the two points ✓

   D. The diagonals of the plane

4. **Given two points A(2, 3) and B(6, 7), what is the distance between them?**

   A. 4

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B. 5

C. 32\sqrt{32} ✓

D. 16\sqrt{16}

5. **Why is the Pythagorean Theorem useful in data science?**

A. To estimate population growth

B. To find the midpoint of a data range

C. To calculate the distance between data points in spatial or clustering problems ✓

D. To convert data from one unit to another

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KM-02-KT07: Operator precedence

1. **What is operator precedence in programming or mathematics? Please provide an example to illustrate how it affects the evaluation of an expression.**

Operator precedence is a set of rules that determines the order in which operations are performed in an expression. For example, in the expression 5 + 3 * 2, multiplication has higher precedence than addition. So, 3 * 2 is evaluated first (resulting in 6), and then 5 + 6 is calculated, giving a final result of **11**. Without precedence, a left-to-right evaluation would give 5 + 3 = 8, then 8 * 2 = 16, which is incorrect.

2. **Why are operator precedence and associativity important when evaluating expressions in mathematics or programming?**

They are crucial to ensure that expressions are evaluated **correctly, consistently, and unambiguously**, regardless of who (or which computer) is evaluating them. This prevents errors and ensures that the intended mathematical logic is preserved.

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Self-Check Quiz: Operator Precedence

Instructions: Choose the best answer for each question. These questions assess your understanding of operator precedence, order of operations, and how they apply in programming and mathematical expressions.

1. **Which term best describes "operator precedence"?**

   A. The rule that determines variable names

   B. The order in which operations are evaluated in an expression ✓

   C. The order in which variables are declared

   D. The order of values in a dataset

2. **Why is operator precedence important in both math and programming?**

   A. To keep the code short

   B. To reduce the number of variables

   C. To ensure expressions are evaluated correctly and consistently ✓

   D. To avoid using parentheses

3. **Which of the following expressions evaluates to 14?**

   A. 2 + 3 * 4

   B. (2 + 3) * 4

   C. 2 + (3 * 4) ✗

   D. 2 * 3 + 4

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4. **In the expression:**

8 + 4 * 2 - 3

What is the correct order of operations?

A. Add, Multiply, Subtract

B. Multiply, Add, Subtract ✓

C. Multiply, Subtract, Add

D. Add, Subtract, Multiply

5. **According to PEDMAS, which operation should be performed first?**

A. Addition

B. Multiplication

C. Parentheses ✓

D. Division

6. **What is the result of this expression in most programming languages?**

5 + 6 * 2

A. 22

B. 17 ✓

C. 11

D. 28

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7. **Which of the following best describes how PEDMAS is applied in programming?**

A. All operations are evaluated left to right

B. Only parentheses are used

C. The same rules apply as in math: Parentheses → Exponents → Multiplication/Division → Addition/Subtraction ✓

D. Operators are executed randomly depending on the language

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KM-02-KT08: Integer division

1. **What role does division play in data analysis and problem solving?**

Division is fundamental for calculating averages, rates, proportions, percentages, and for normalizing data. It helps in comparing parts of a dataset to the whole and is essential for creating meaningful metrics and KPIs (Key Performance Indicators).

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2. Divide 10 by 4, and explain how integer division discards the fractional part of a result.

10 ÷ 4 = 2.5. In integer division, the fractional part (the .5) is simply discarded (or truncated), leaving only the whole number quotient, which is **2**.

3. What is the rule for dividing integers, and how does the sign of the result depend on the signs of the numbers?

- **Positive** if both numbers have the same sign (Positive ÷ Positive, Negative ÷ Negative).
- **Negative** if the two numbers have different signs (Positive ÷ Negative, Negative ÷ Positive).

Self-Check Quiz: Dividing Integers

Instructions: Highlight the most appropriate answer for each question. You may use scrap paper for calculations.

1. **What is the result of this division?**

20 ÷ 4

A. 4

B. 5

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C. 6

D. 8

2. **Which of the following is the correct result of dividing a negative number by a positive number?**

-15 ÷ 3

A. -5 ✓

B. 5

C. -3

D. 0

3. **What is the sign of the result when you divide two integers with the same sign?**

A. Always negative

B. Always positive ✓

C. Depends on the numbers

D. Zero

4. **What is the result of this operation?**

-24 ÷ -6

A. -4

B. 4 ✓

C. 6

D. -6

5. **Which of the following statements is TRUE?**

A. Division of integers always results in a whole number

B. Division by zero is undefined ✓

C. The result of division is always smaller than the dividend

D. Negative ÷ Negative = Negative

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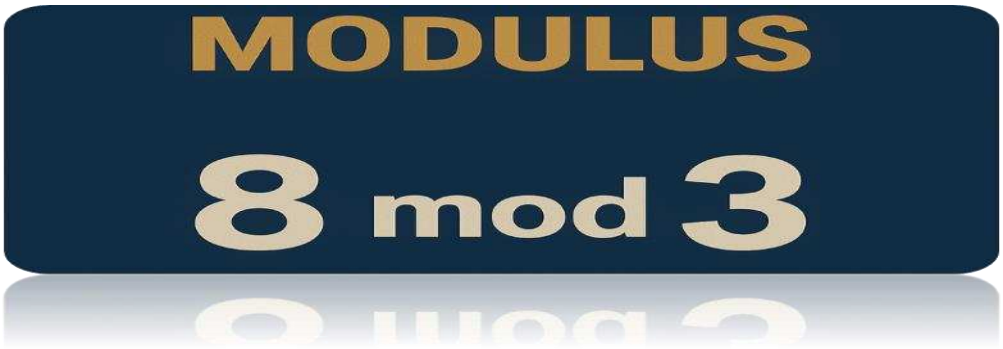

KM-02-KT09: Modulus

1. **Explain how the modulus rule is applied in this example: 17 % 5**

The modulus operator (%) returns the remainder after integer division. For 17 % 5, we see how many times 5 fits into 17. 5 * 3 = 15. The remainder is 17 - 15 = 2. Therefore, 17 % 5 = 2.

Self-Check Quiz: Application of the Modulus Operator (%)

Instructions: Highlight the correct answer for each question. The modulus operator returns the remainder after division of two integers.

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1. **What is the result of the following operation?**

10 % 3

A. 1 ✓

B. 3

C. 0

D. 2

2. **What does the expression 15 % 5 evaluate to?**

A. 0 ✓

B. 5

C. 3

D. 10

3. **Which use case is appropriate for applying the modulus operator?**

A. Finding the average of a dataset

B. Determining whether a number is even or odd ✓

C. Sorting a list in order

D. Multiplying percentages

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4. **What is the result of -13 % 5 in most programming languages?**

   A. -3 ✓

   B. 2

   C. -2

   D. 3

5. **In data science, how might the modulus operator be practically applied?**

   A. To check if a value belongs to a string

   B. To loop through rows of a dataset

   C. To filter every nth row or to alternate row operations ✓

   D. To replace missing data with the median

Self-Check Quiz: Application of the Equality Operator (==)

Instructions: Highlight the most appropriate answer for each question. Remember, == checks whether two values are equal and returns a Boolean result: True or False.

1. **What does the operator == do in programming?**

   A. Assigns a value to a variable

   B. Compares two values for equality ✓

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C. Checks if a value is not equal

D. Increments a variable

2. **What is the result of this expression?**

7 == 5 + 2

A. True ✓

B. False

C. 7

D. Error

3. **In Python, what will be the output of this code?**

a = 10

b = 20

a == b

A. 10

B. 20

C. True

D. False ✓

4. **Which of the following comparisons will return True?**

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A. 4 == "4"

B. 3 == 3.0 ✓

C. True == "True"

D. "data" == "Data"

5. **How is the == operator commonly used in data science programming?**

A. To create new variables

B. To compare two datasets or values (e.g., for filtering rows) ✓

C. To format strings

D. To plot data

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KM-02-KT10: Increments

1. **Explain the concepts of increments with an example.**

An increment is the process of increasing a value by a fixed amount, usually 1. For example, the increment operation x = x + 1 or x++ increases the value of variable x by 1. If x was 5, after the increment it becomes 6.

2. **What are compound assignment operators with an example?**

Compound assignment operators perform an operation and an assignment in one step.

- Example: x += 5 is shorthand for x = x + 5.

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3. **Use the example y *= 3 to explain the concept of a compound assignment operator.**

The expression y *= 3 is a compound assignment operator. It means "take the current value of y, multiply it by 3, and then assign the result back to y". It is equivalent to writing y = y * 3. If y was 4, after y *= 3, its new value would be 12.

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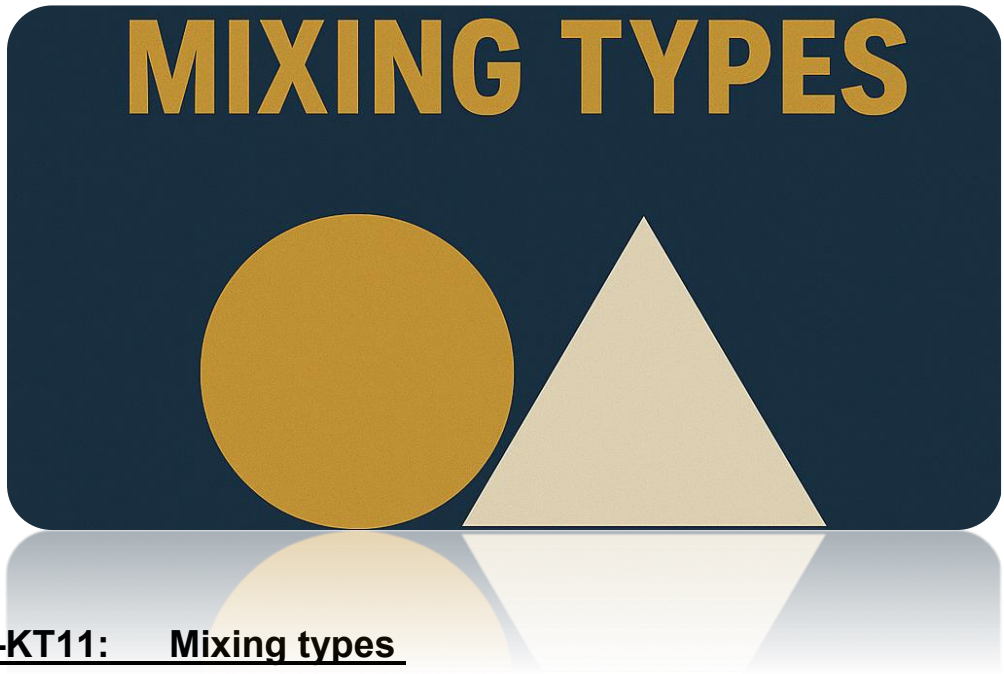

KM-02-KT11: Mixing types

Self-Check Quiz: Mixing Types and Expression Evaluation

Instructions: Answer the following questions. Use your knowledge of operator precedence, expression order, and how mixed types behave in calculations. Highlight the correct answer.

1. **What is the result and type of the following expression?**

5 + 3.0 * 2

A. 16 (Integer)

B. 11.0 (Float) ✓

C. 11 (Integer)

D. 8.0 (Float)

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2. **Which expression gives a different result than the others?**

A. 10 + 4 / 2.0

B. (10 + 4) / 2.0 ✓

C. 10 + (4 / 2.0)

3. **What will be the type of the result in this expression?**

result = 7 / 2

A. Integer

B. Float ✓

C. String

D. Error

4. **Which of the following best explains what the computer does in the expression x += 2.5, where x is initially an integer?**

A. It adds 2.5 to x and converts the result to an integer

B. It gives an error

C. It adds 2.5 to x and stores the result as a float ✓

D. It multiplies x by 2.5

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5. **What will the final value of x be in the following?**

x = 5

x *= 2.0

A. 10

B. 10.0 ✓

C. 7.0

D. Error

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